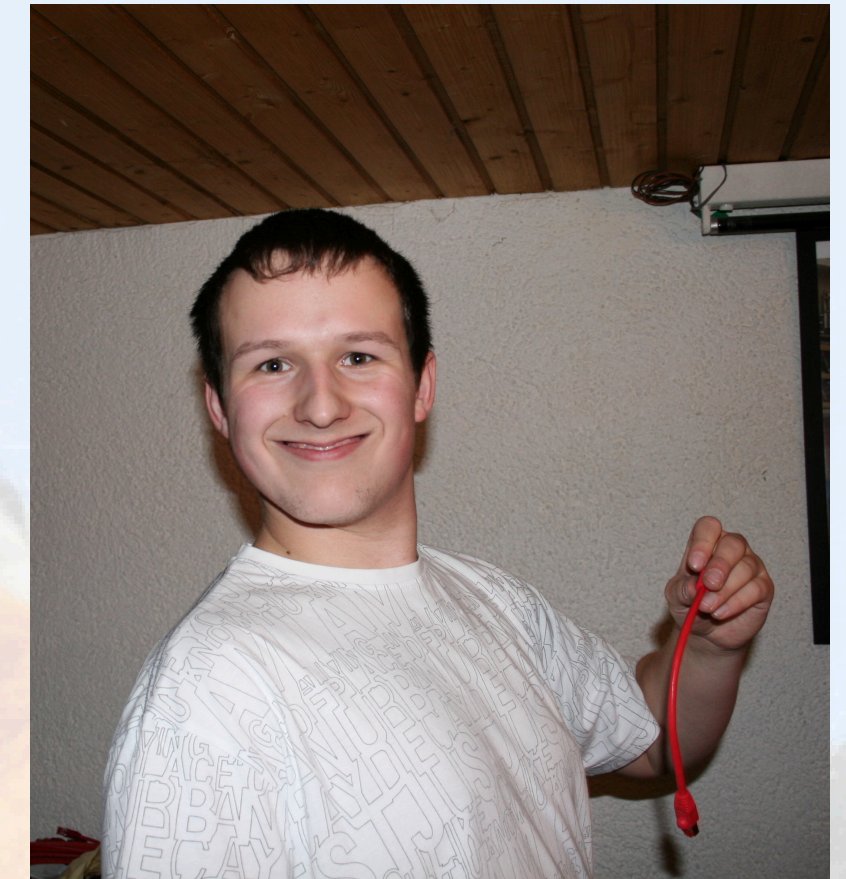


Foreman + OpenBolt

a ♥ story in three acts

\$ whoami

- Tim 'bastelfreak' Meusel
- Puppet Contributor since 2012
- Merging stuff on [Vox Pupuli](#) since 2015
- Vox Pupuli PMC member
- Principal IT Automation Consultant at [betadots](#)









OpenBolt

- Runs tasks via ssh or WinRM on remote systems
 - CLI only
- A task is a executable binary/script + json file
- Also supports Podman/Docker/Jails/lxd targets



OpenBolt

```
{
  "puppet_task_version": 1,
  "supports_noop": false,
  "description": "Switch feat branch to production",
  "parameters": {
    "control_repo_branch": {
      "description": "Control-repo branch",
      "type": "String[1]"
    }
  }
}
```

```
#!/bin/bash
```

```
set -euo pipefail
```

```
pushd /tmp/control-repo || exit
git branch --move production old_prod
git branch --move "$PT_control_repo_branch" production
```



OpenBolt

- Runs tasks via ssh or WinRM on remote systems
 - CLI only
- A task is a executable binary/script + json file
- Also supports Podman/Docker/Jails/lxd targets
- Input and output of each task is JSON
 - makes parsing, scripting and concatenating easy



Convert the PE console into a foreman plugin #28

 Open



bastelfreak opened on Apr 16, 2024



Use Case

Foreman is an established open source tool for life cycle management. It has a working open source community, different companies offer commercial support/training/consulting for it. There are also commercial downstream products like RedHat Satellite or [Orcharhino](#). Foreman is used for many on premise setups, it has a proper CI setup and a plugin setup. It already has support to view puppet reports and facts. With [HDM](#) it has also an option to view, edit and diff Hiera data. Foreman can be used as an ENC and as hiera backend. And it has support for run remote jobs.

Foreman and the PE console overlap a lot in their feature set. I think Foreman is the better implementation because it looks more healthy. There are more docs, in general more activity and people can contribute to it. Contribute to the PE console is impossible and feature requests are mostly ignored.

I think it makes sense to replace the standalone PE console with a set of foreman plugins. Even if those are commercial/closed source people can at least contribute fixes to the rest of foreman and I assume that maintenance of the whole PE console codebase requires more time than it takes to maintain a small set of foreman plugins.

And since many setups already use Satellite/Orcharhino/Foreman they are already used to the UI and they don't need a second Web UI.

github.com/puppetlabs/puppet-enterprise_issues/issues/28

RFC: Integrating bolt into remote execution

■ Development ■ RFCs puppet remoteexecution rfc



bastelfreak

1 Oct 2024

Hi!

I would like to add support for bolt tasks to the remote execution architecture in foreman.

What's bolt?

An open source tool from same people as Puppet ([Bolt Documentation](#) ¹⁰). It supports running commands (shell or powershell), scripts (anything that's executable), tasks or plans.

What's a task?

A task is anything that's executable + a json file describing the purpose of a task and optional input parameters with datatypes. Tasks return json. bolt has a CLI interface and an API, no web UI. bolt can use different transport protocols, ssh and winrm are available by default. It has a plugin system.

Rationale:

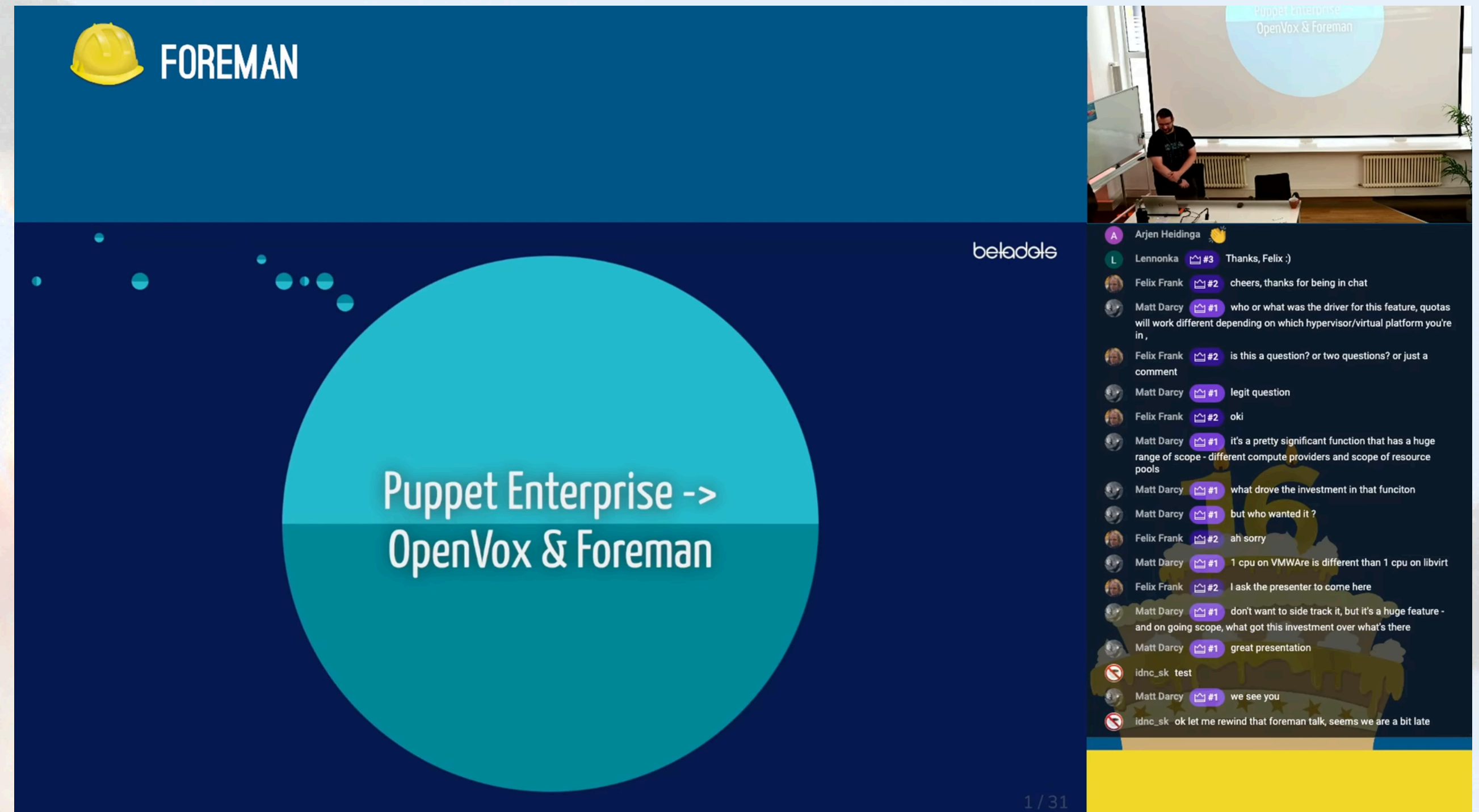
Puppet Enterprise (costs per node per year) has a Web UI to start tasks. I think it would be a great fit for the foreman UI to also support bolt tasks. bolt tasks are heavily used within the Puppet Enterprise and Puppet Opensource ecosystem for orchestration. For example OS patching whole clusters or doing software rollouts.

community.theforeman.org/t/rfc-integrating-bolt-into-remote-execution

OpenBolt

How it started

Act 1



The screenshot shows a presentation slide with a dark blue background. At the top left, there is a yellow hard hat icon followed by the word 'FOREMAN' in white. In the center, a large teal circle contains the text 'Puppet Enterprise -> OpenVox & Foreman' in white. The bottom right corner of the slide displays '1 / 31'. On the right side, there is a vertical chat window with a dark background and white text. The chat header includes a small video feed of a presenter and the text 'Puppet Enterprise OpenVox & Foreman'. The chat messages are as follows:

- Arjen Heidinga: [reaction]
- Lennonka: [reaction] #3 Thanks, Felix :)
- Felix Frank: [reaction] #2 cheers, thanks for being in chat
- Matt Darcy: [reaction] #1 who or what was the driver for this feature, quotas will work different depending on which hypervisor/virtual platform you're in ,
- Felix Frank: [reaction] #2 is this a question? or two questions? or just a comment
- Matt Darcy: [reaction] #1 legit question
- Felix Frank: [reaction] #2 oki
- Matt Darcy: [reaction] #1 it's a pretty significant function that has a huge range of scope - different compute providers and scope of resource pools
- Matt Darcy: [reaction] #1 what drove the investment in that function
- Matt Darcy: [reaction] #1 but who wanted it ?
- Felix Frank: [reaction] #2 ah sorry
- Matt Darcy: [reaction] #1 1 cpu on VMWare is different than 1 cpu on libvirt
- Felix Frank: [reaction] #2 I ask the presenter to come here
- Matt Darcy: [reaction] #1 don't want to side track it, but it's a huge feature - and on going scope, what got this investment over what's there
- Matt Darcy: [reaction] #1 great presentation
- idnc_sk: test
- Matt Darcy: [reaction] #1 we see you
- idnc_sk: ok let me rewind that foreman talk, seems we are a bit late

- Recording: youtu.be/NRtstDVtpyk?t=21810



OpenBolt

How it started

Act 1

Act 2



bastelfreak

Jan 30

Hey people, I will also do a live demo at the Config Management camp in Ghent next week:

 cfp.cfgmgmtcamp.org



**Bringing OpenBolt into Foreman CfgMgmtCamp 2026
Ghent** ⁷

[OpenBolt](<https://github.com/OpenVoxProject/openbolt>) is a CLI application from the [OpenVoxProject](<https://voxpupuli.org/openvox/>). It provides different ways to run your beloved sysadmin scripts with proper guardrails and it makes it easy to...

Hit me up if you have questions or thoughts!

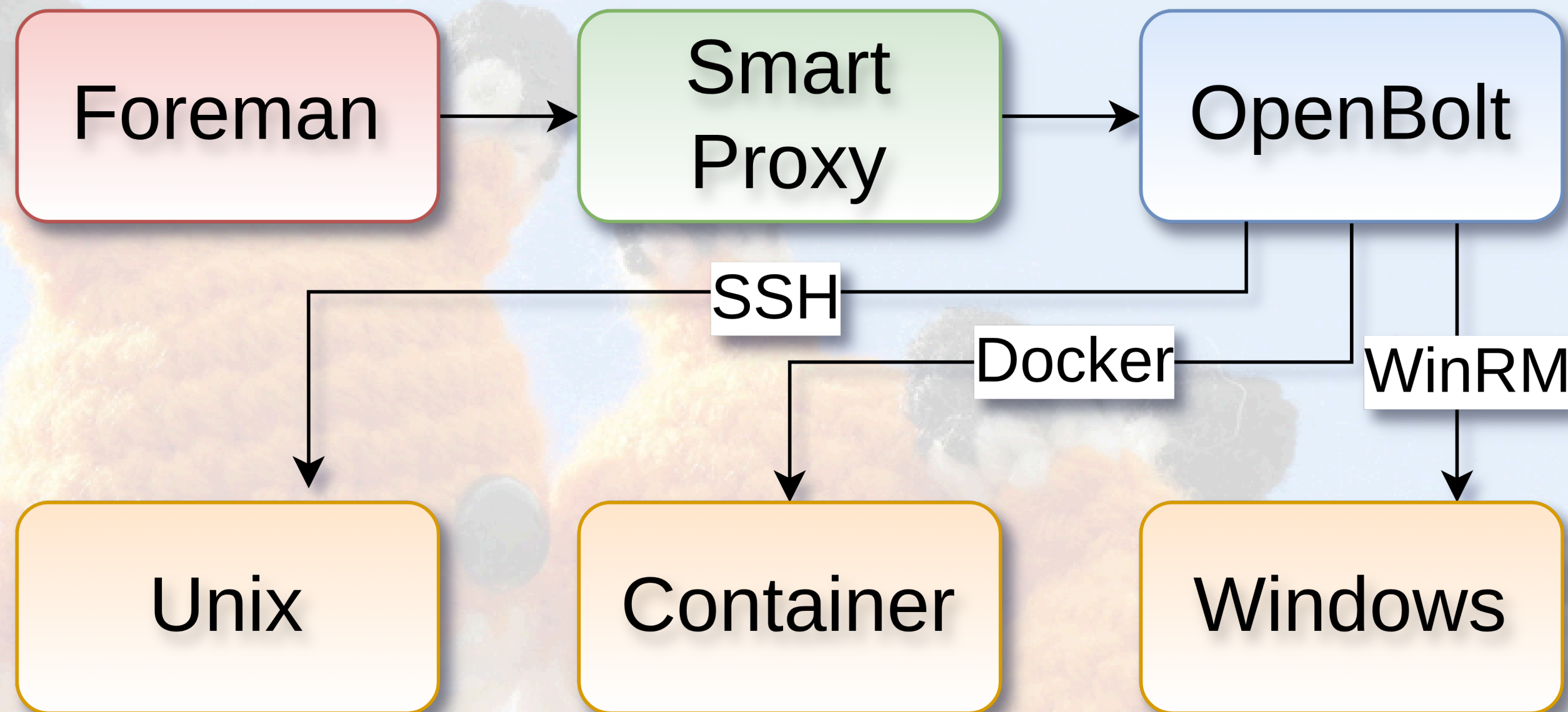
- Slideset: bastelfreak.de/cfgmgmtcamp2026/openbolt.html

OpenBolt

How it started

Act 1

Act 2



OpenBolt

How it started

Act 1

Act 2

Client

Broker

Server



Persistent mTLS



Transcient mTLS

- Choria Orchestration System

OpenBolt

How it started

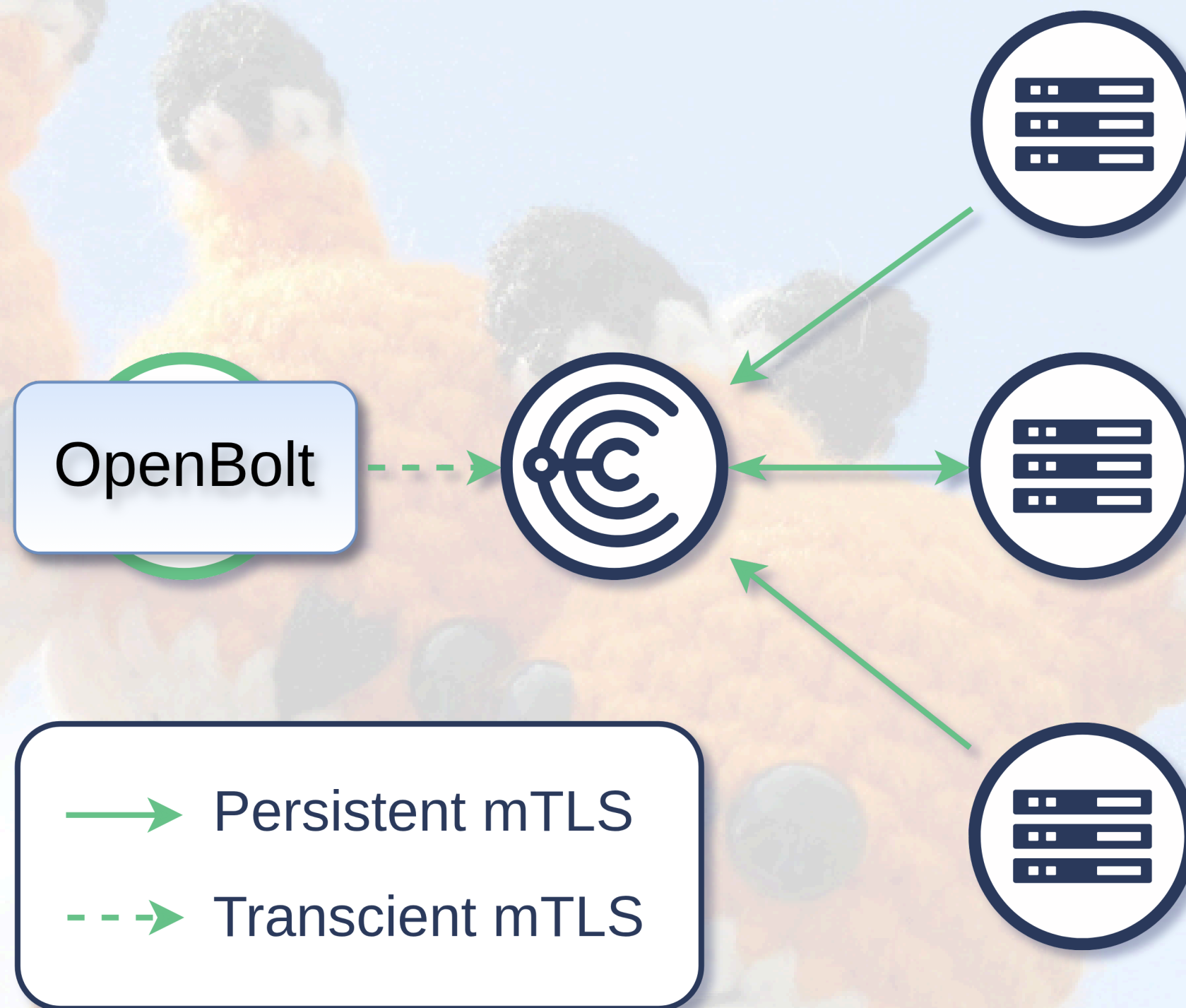
Act 1

Act 2

Client

Broker

Server



- Choria Orchestration System

Act 3

OpenBolt

- github.com/overlookinfra/foreman_openbolt - OpenBolt Plugin for Foreman

How it started

Act 1

Act 2

Act 3



OpenBolt

How it started

Act 1

Act 2

Act 3

- github.com/overlookinfra/foreman_openbolt - OpenBolt Plugin for Foreman
- github.com/overlookinfra/smart_proxy_openbolt - OpenBolt Smartproxy Plugin



OpenBolt

How it started

Act 1

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Act 3

- github.com/overlookinfra/foreman_openbolt - OpenBolt Plugin for Foreman
- github.com/overlookinfra/smart_proxy_openbolt - OpenBolt Smartproxy Plugin
- github.com/theforeman/foreman-installer/pull/1049 - foreman-installer Integration



OpenBolt

How it started

Act 1

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Act 3

- github.com/overlookinfra/foreman_openbolt - OpenBolt Plugin for Foreman
- github.com/overlookinfra/smart_proxy_openbolt - OpenBolt Smartproxy Plugin
- github.com/theforeman/foreman-installer/pull/1049 - foreman-installer Integration
- github.com/OpenVoxProject/openbolt/pull/206 - Integrate Choria Transport into OpenBolt



OpenBolt

How it started

Act 1

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Act 3

- github.com/overlookinfra/foreman_openbolt - OpenBolt Plugin for Foreman
- github.com/overlookinfra/smart_proxy_openbolt - OpenBolt Smartproxy Plugin
- github.com/theforeman/foreman-installer/pull/1049 - foreman-installer Integration
- github.com/OpenVoxProject/openbolt/pull/206 - Integrate Choria Transport into OpenBolt
- All changes are merged & released & packaged for Foreman 3.17.1+



OpenBolt

How it started

Act 1

Act 2

Act 3

 **FOREMAN**

Default Organization ▾

Default Location ▾

Q Search and go

 Monitor >

 Hosts >

 OpenBolt >

 Configure >

 Infrastructure >

 Administer >

Overview

Q Search

➔



Latest Jobs

No jobs available



OpenBolt

How it started

Act 1

Act 2

Act 3

Launch OpenBolt Task

 Launch Task

Smart Proxy

foreman-smartproxy.local

Hosts

▼ Hosts

Filter by hosts

Task Name

Select Task

↺

Parameters



Select a task to see parameters

OpenBolt Options

	Name	Value
›	transport	sshwinrmchoria
›	host-key-check	☒ host-key-check
›	noop	☐ noop
›	verbose	☐ verbose
›	log-level	debug
›	password	
›	private-key	
›	run-as	
›	sudo-password	
›	tmpdir	
›	user	



OpenBolt

How it started

Act 1

Act 2

Act 3

Task Name

service

Parameters

	Name	Value
>	action *	
>	name *	
>	force	☐
>	provider	



OpenBolt

How it started

Act 1

Act 2

Act 3

Task Name

service

Parameters

Name	Value
▼ action *	
Type: Enum[start, stop, restart, enable, disable, status] This field is required The operation (start, stop, restart, enable, disable, status) to perform on the service.	
▼ name *	
Type: String[1] This field is required The name of the service to operate on.	
▼ force	☐
Type: Optional[Boolean] Force a Windows service to restart even if it has dependent services. This parameter is passed for Windows services only.	
▼ provider	
Type: Optional[String[1]] The provider to use to manage or inspect the service, defaults to the system service manager. Only used when the 'puppet-agent' feature is available on the target so we can leverage Puppet.	



OpenBolt

How it started

Act 1

Act 2

Act 3

Task Execution

Run Another Task



Execution Details



Task Details

Smart Proxy	foreman-smartproxy.local
Job ID	4db2e7c2-ad22-404c-942f-458ec594c7b1
Hosts	2
Status	 Running Updating every 5 seconds...
Execution Time	10s



OpenBolt

How it started

Act 1

Act 2

Act 3

Task Execution

Run Another Task

Execution Details Task Details

Smart Proxy	foreman-smartproxy.local
Job ID	6d74c10b-f5a4-47d3-92dc-781620c299c5
Hosts	1
Status	Failed
Execution Time	5s

Result Log Output

```
{
  "items": [
    {
      "value": {
        "_error": {
          "msg": "Host key verification failed for testnode.local: fingerprint SHA256:dIVGoRGJUxLRsoj+pdqpXvnE5Li7YRfsRV4i3SEGavY is unknown for \"foreman-smartproxy.local,127.0.1.1\\",
          "kind": "puppetlabs.tasks/connect-error",
          "details": {},
          "issue_code": "HOST_KEY_ERROR"
        }
      },
      "action": "task",
      "object": null,
      "status": "failure",
      "target": "foreman-smartproxy.local"
    }
  ],
  "elapsed_time": 0,
  "target_count": 1
}
```



OpenBolt

How it started

Act 1

Act 2

Act 3



Result



Log Output



Copy

```
OpenBolt command: bolt task run service --targets foreman-smartproxy.local --no-save-rerun --  
concurrency=100 --connect-timeout=30 --project /etc/puppetlabs/code/environments/production -  
-format json --no-color --transport=ssh force=false action=status name=puppet
```

OpenBolt

How it started

Act 1

Act 2

Act 3

Result

Log Output

Copy

```
{
  "items": [
    {
      "value": {
        "_error": {
          "msg": "Host key verification failed for testnode.local: fingerprint SHA256:dIVGoRGJUXLRsoj+pdqpXvnE5Li7YRfsRV4i3SEGavY is unknown for \"foreman-smartproxy.local,127.0.1.1\"",
          "kind": "puppetlabs.tasks/connect-error",
          "details": {},
          "issue_code": "HOST_KEY_ERROR"
        }
      },
      "action": "task",
      "object": null,
      "status": "failure",
      "target": "foreman-smartproxy.local"
    }
  ],
  "elapsed_time": 0,
  "target_count": 1
}
```



OpenBolt

How it started

Act 1

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Act 3

Execution Details

Task Details

Smart Proxy

[foreman-smartproxy.local](#)

Job ID

4db2e7c2-ad22-404c-942f-458ec594c7b1

Hosts

2

Status

Success

Execution Time

1m6s

Result

Log Output

```
{
  "items": [
    {
      "value": {
        "status": "MainPID=20148,LoadState=loaded,ActiveState=active",
        "enabled": "enabled"
      },
      "action": "task",
      "object": "service",
      "status": "success",
      "target": "foreman.local"
    },
    {
      "value": {
        "status": "MainPID=103902,LoadState=loaded,ActiveState=active",
        "enabled": "enabled"
      },
      "action": "task",
      "object": "service",
      "status": "success",
      "target": "puppet.local"
    }
  ],
  "elapsed_time": 61,
  "target_count": 2
}
```



OpenBolt

How it started

Act 1

Act 2

Act 3

Conclusion

- OpenBolt is a great addition to the Foreman ecosystem
 - Makes it easy to reuse existing Tasks (and soon Plans)



OpenBolt

How it started

Act 1

Act 2

Act 3

Conclusion

- OpenBolt is a great addition to the Foreman ecosystem
 - Makes it easy to reuse existing Tasks (and soon Plans)
- Choria provides almost infinite scaling and opposite firewalling



OpenBolt

How it started

Act 1

Act 2

Act 3

Conclusion

- OpenBolt is a great addition to the Foreman ecosystem
 - Makes it easy to reuse existing Tasks (and soon Plans)
- Choria provides almost infinite scaling and opposite firewalling
- Please join our [IRC/Slack](#) to provide feedback
- For feedback: bastelfreak on IRC/ Slack or tim@bastelfreak.de
- This talk and previous ones: github.com/bastelfreak/talks

Thanks for your attention!



OpenBolt

How it started

Act 1

Act 2

Act 3

Conclusion

Act 4

```
# @summary one plan to rule them all!
# @param version which Foreman version do we want?
# @param node systemd where we install foreman
plan foreman::install (
  String[1] $version = '3.19.1',
  TargetSpec $node,
) {
  $os = run_task('facts', $node).first['os']['family']
  if $os == 'RedHat' {
    run_task('yum::update', $node)
  }
  out::message("setting up foreman on ${node}")
  run_task('foreman::add_repo', $node)
  $package = {'action' => 'install', 'name' => 'foreman-service'}
  run_task('package', $node, $package)
  $service = {'action' => 'start', 'name' => 'foreman'}
  $result = run_task('service', $node, $service)
  if $result.ok {
    out::message("worked on ${result.ok_set}")
  }
}
```